

Unit 4.5 Curve Sketching

Note Title

2015/05/12

$$f. \begin{cases} D - \text{domain} \\ I - \text{intercepts } (x \text{ \& } y) \\ S - \text{symmetry (even/odd/neither)} \\ A - \text{asymptotes (vertical/horizontal)} \end{cases}$$

$$f' \begin{cases} D - \text{decreasing/increasing} \\ E - \text{extremes} \end{cases}$$

$$f'' \begin{cases} C - \text{concavity} \\ I - \text{inflection points} \end{cases}$$

1. Sketch $f(x) = xe^{-x}$

$$\text{Given } \begin{cases} f'(x) = (1-x)e^{-x} \\ f''(x) = (x-2)e^{-x} \end{cases}$$

D : $\mathbb{R}$

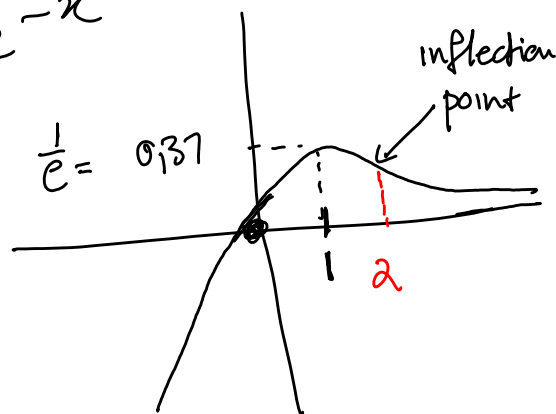
$$I : \begin{aligned} x=0 &\Rightarrow y=0 \\ y=0 &\Rightarrow x=0 \end{aligned}$$

$$S : f(-x) = -xe^x \neq f(x) \neq -f(x)$$

Not even / not odd

$$\begin{aligned} A : \lim_{x \rightarrow \infty} xe^{-x} & \quad (\infty \times 0) \\ &= \lim_{x \rightarrow \infty} \frac{x}{e^x} \quad \left(\frac{\infty}{\infty} \right) \\ &= \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0 \end{aligned}$$

$y=0$ is a HA



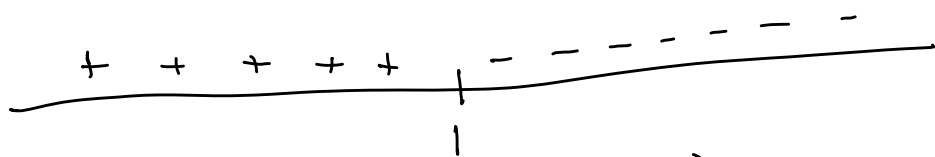
HA.

$$\lim_{x \rightarrow -\infty} x e^{-x} = -\infty \quad (-\infty \times \infty)$$

VA: $f(x) = \frac{x}{e^x}$ No VA's.

D: $f'(x) = (1-x)e^{-x}$

CP: $f' = 0 \Rightarrow x = 1$,

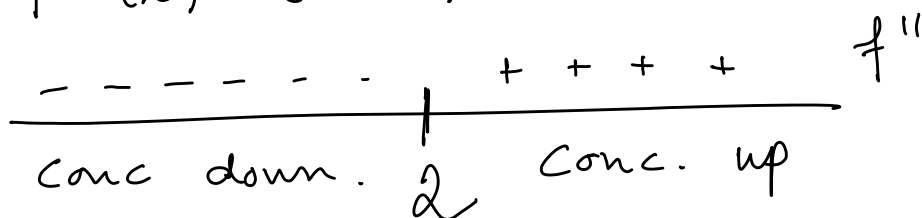


f increases on $(-\infty, 1)$

f decreases on $(1, \infty)$

E: f has a local max $(1, e^{-1}) \approx (1, 0.37)$

C: $f''(x) = (x-2)e^{-x}$



I: f has an inflection point in $x=2$.

Example:

Sketch $f(x) = \frac{x^2}{x-2}$, given that

$$f'(x) = \frac{x(x-4)}{(x-2)^2}, \quad f''(x) = \frac{8}{(x-2)^3}$$

D: $\mathbb{R} \setminus \{2\}$

I: $x=0 \Rightarrow y=0$
 $y=0 \Rightarrow x=0$

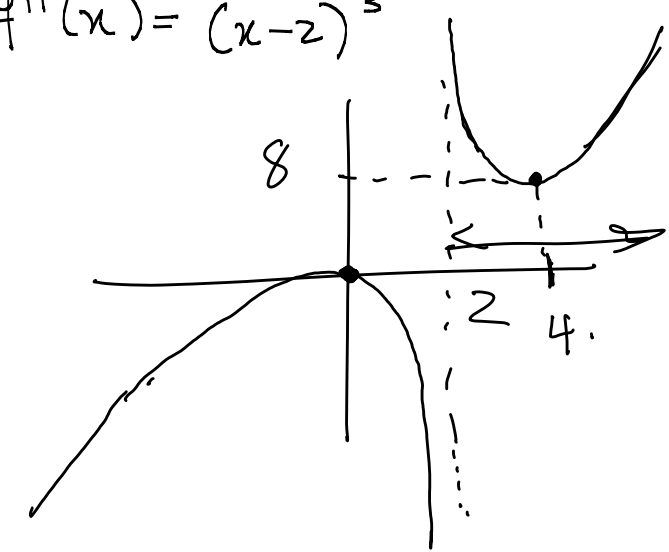
S: Not even / not odd.

A: HA: $\lim_{x \rightarrow \infty} \frac{x^2}{x-2} = \lim_{x \rightarrow \infty} \frac{x}{1 - \frac{2}{x}} = \infty$
 $\lim_{x \rightarrow -\infty} \frac{x}{1 - \frac{2}{x}} = -\infty$.

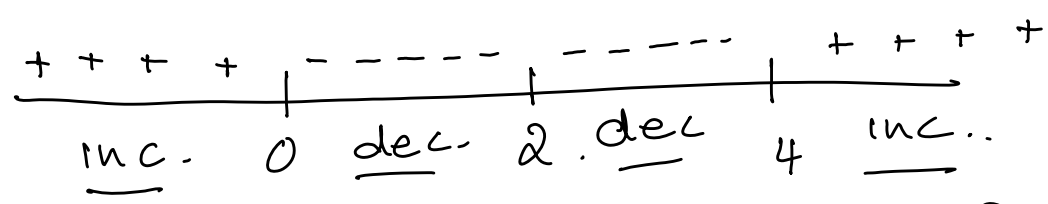
VA: $x=2$ is a VA.

$$\lim_{x \rightarrow 2^+} \frac{x^2}{x-2} = \infty$$

$$\lim_{x \rightarrow 2^-} \frac{x^2}{x-2} = -\infty$$



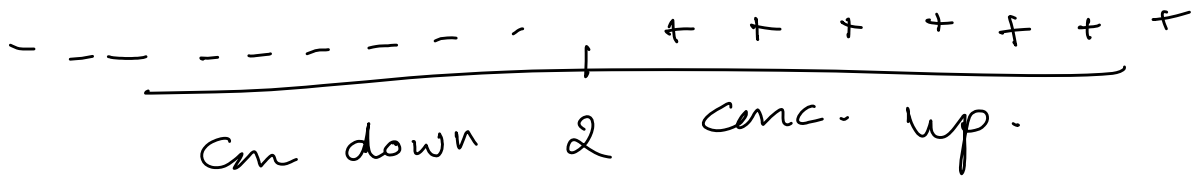
D: $f'(x) = \frac{x(x-4)}{(x-2)^2}$



E: f has a local max $(0, 0)$
 f has a local min $(4, 8)$

C:

$$f''(x) = \frac{8}{(x-2)^3}$$



I: No inflection points.

Example

$$f(x) = \frac{x}{x^2+1}, \quad f'(x) = \frac{1-x^2}{(x^2+1)^2}$$

$$f''(x) = \frac{2x(x^2-3)}{(x^2+1)^3}$$

D: $\mathbb{R}$

I: $(0,0)$

S: Odd function

A: No VA.

$$\lim_{x \rightarrow \infty} f(x) = 0, \quad \lim_{x \rightarrow -\infty} f(x) = 0$$

$y=0$ is a HA.

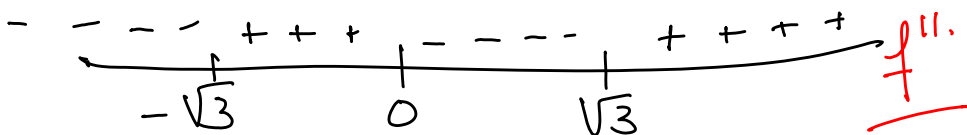
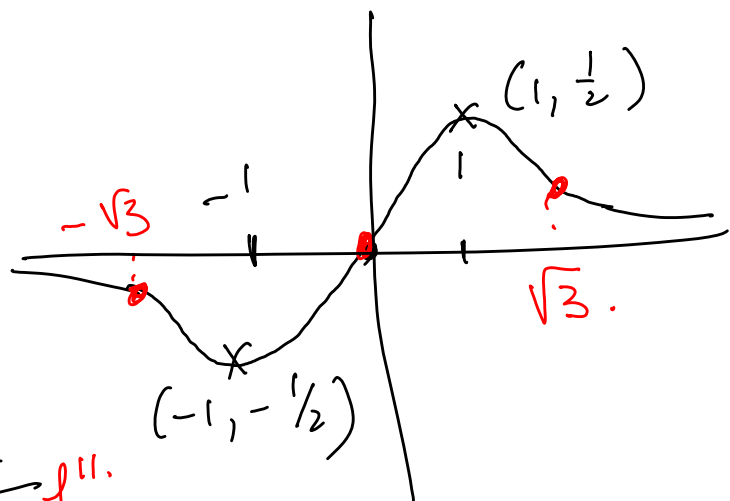
$$D: f'(x) = 0 \Rightarrow x = \pm 1$$

f increases on $(-1, 1)$ and
 f decreases on $(-\infty, -1) \cup (1, \infty)$.

E: Local min $(-1, -\frac{1}{2})$
 Local max $(1, \frac{1}{2})$.

$$C: f'' = \frac{2x(x^2 - 3)}{(x^2 + 1)^3}$$

$$f'' = 0 \Rightarrow x = 0, \quad x = \pm\sqrt{3}$$



I: Points of inflection: $x = -\sqrt{3}$
 $x = 0$
 $x = \sqrt{3}$.

Example: Sketch $f(x) = 12x^{1/2} - 2x^{3/2}$
 $(= 2\sqrt{x}(6-x))$

Given $f'(x) = 3x^{-1/2}(2-x)$

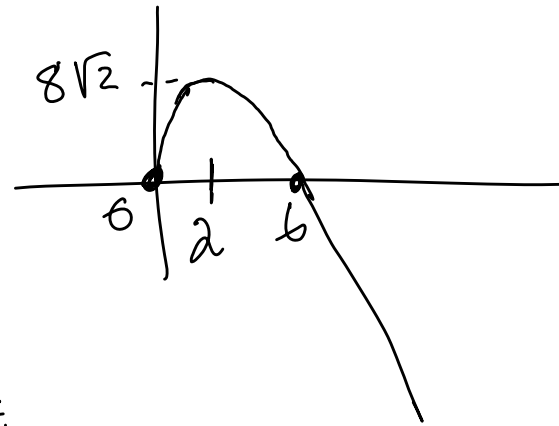
$f''(x) = -\frac{3}{2}x^{-3/2}(2+x)$

D: $[0, \infty)$

I: $(0,0)$, $(6,0)$.

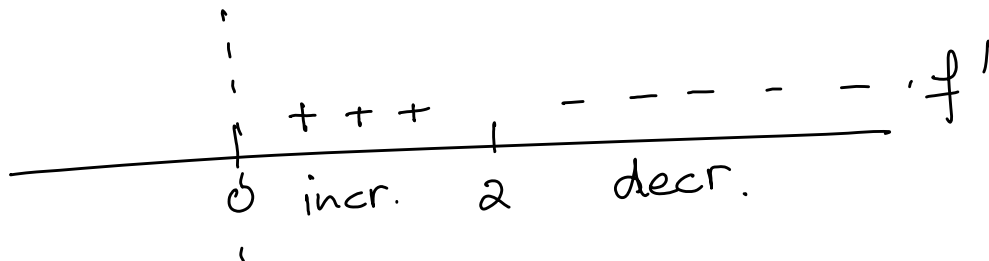
S: Not applicable

A: No VA, No HA, $\lim_{x \rightarrow \infty} f(x) = -\infty$.



D: $f'(x) = 0 \Rightarrow x = 2$

$x=0$ is also a critical point



f decreases on $(2, \infty)$

f increases on $(0, 2)$

E: Local max $(2, 8\sqrt{2})$

C: $f'' < 0$ for $x \in (0, \infty)$
 f is concave down on $[0, \infty)$

I: No inflection points.